

Diesel engine

12V4000M70

Spare parts catalog for product no.

526101324

Printed in Germany

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Subject to alterations and amendments.

Preface

This catalog is published for parts requisitioning purposes and provides assistance in engine disassembly and reassembly. It contains a complete list of all engine components, including those which are not available as replacement parts.

The parts originate from the Engine Design File and document the product's construction status.

Catalog Arrangement

The catalog is subdivided into functional groups and functional group segments (FG/FGS).

The headings of the illustration pages contain the following information:

- ◆ TYPE
- ◆ SERIAL NO.
- ◆ FG/FGS DESTINATION
- ◆ FG/FGS NO.
- ◆ TYP

Explanation of Columns

- ◆ POS: Provides a cross-reference from illustration to text section. Non-illustrated parts are indicated by the suffix "Z" after the C/N.
- ◆ PART NO.: This is the valid number for ordering the associated part.
- ◆ DESIGNATION: The precise part name.
- ◆ QTY. (Quantity of Unit): This column indicates the quantity of this part required for the functional group.
- ◆ MEASUREMENT
- ◆ FN (Footnote): Provides additional parts information. For clarification of footnote symbols, see Page 3/2.
- ◆ REMARKS: This column is intended for supplementary information.

Amendments Service

Part No. changes and amendments which occur after publication of the catalog will not be published. We therefore recommend updating of the catalog part numbers by the end user. This information is provided, for instance, in order confirmations and/or on equipment labels.

Footnote Symbols Identification

- A Index footnote
- AA See separate manual
- AB Supplied as a complete assembly. Repair by manufacturer only.
- AC No part number assigned. State illustration table number, callout number, designation and catalog number when ordering.
- AD Quantity referenced to one unit.
- E Magnetically treated part or assembly (stamped)
- EM Emission-relevant parts. In the case of all marine engines with IMO certification it should be noted that identification number of the emission-relevant parts on the engine must agree with the details given in the technical file. Give the EM code also when ordering with the order No., e.g. 535 030 04 17 EM.
- K Following reasons require ordering of complete parts group:
 - KA Part machined jointly with other components
 - KB Inseparable connection
 - KC Machining on group assembly
 - KD Group balancing required
 - KE Individual parts not obtainable through the parts supply system
 - KF Replacement part will be assigned a different part number
 - KG Available separately
- M Parts having different repair sizes.

Dimensions are listed in the Tolerances and Wear Limits List.

NB As required

P Bought-out components

S Welded - Parts welded together to form an assembly

Component parts are not listed as they are not included in the parts supply system

SA Component part of a welded assembly obtainable through the parts-supply system. Replacement requires welding or brazing

SB Group of parts which are welded or bolted together.

U Part whose dimensions are determined on assembly.

UA Indicate dimensions when ordering

UB Supplied as a complete set only

UC Part requires shaping/bending on assembly

WW Optional

X

Y Symbols used for footnotes appearing on the catalog page.

Z

Ordering Procedure

The following information is essential for spare-part order processing:

- ◆ TYPE
- ◆ SERIAL NO.
- ◆ FG/FGS
- ◆ BOM
- ◆ Pos. number
- ◆ Designation
- ◆ Quantity
- ◆ Page

When ordering replacement parts for ancillary equipment such as injection pumps

Phone orders should be confirmed in writing to prevent errors.

Ordering Example:

Sigla: Universidade FAPESP, RMC, PMS, Pós-graduação em Química Orgânica, São Carlos, SP, Brasil
E-mail: rmc@fapesc.usp.br; pms@fapesc.usp.br

Factory Repairs

Parts shipped to MTU for repair should have an engine serial-number identification (tag or label, for example).

Exchange Components (Reman)

Exchange components are parts overhauled by the manufacturer

New component in exchange procedure

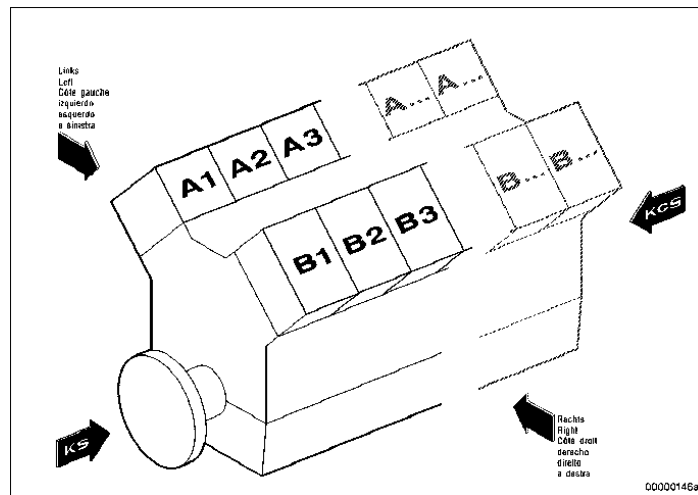
Reman component in exchange procedure

Exchange Procedure

With the introduction of exchange components

For further information refer to Service Information 01/S0028.

Engine Side And Cylinder Designations



Engine sides are always designated as viewed from driving end.

The left bank of cylinders is marked "A" and the right bank "B" (according to DIN ISO 1204).

The cylinders of each bank are numbered consecutively

Numbering of other engine components is also from the drive end

Designations and Abbreviations

Driving end = KS

Free end = KGS

Left side

Right side

Upper side

Lower side

Turbocharger = ATL

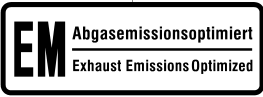
Overview functional groups

FG	FG DESIGNATION
004	LABELS AND MISCELLANEOUS MATERIALS
011	CRANKCASE
013	CYLINDER LINER
014	OIL PAN
015	ENGINE MOUNTING
016	END HOUSING, MAIN PTO END AND AUX PTO END
018	CRANKCASE BREATHER
024	GEARCASE, AUX PTO END
025	GEAR TRAIN, AUX PTO END
031	CRANKSHAFT ...
032	CRANKSHAFT SUPPORT BEARINGS, MAIN PTO END
035	CRANKSHAFT, PTO, AUX PTO END
036	CRANKSHAFT TRANSPORTATION LOCKING DEVICE
037	PISTON
038	CONROD
041	CYLINDER HEAD
042	CYLINDER HEAD ATTACHMENTS
051	CAMSHAFT
052	CAMSHAFT DRIVE
053	CAMSHAFT AXIAL ALIGNMENT BEARING
054	ROLLER TAPPETS, PUSH RODS
055	VALVE DRIVE
073	INJECTION PUMP / HP PUMP
075	FUEL INJECTOR
077	HP FUEL LINE / HP LINE
081	FUEL PIPEWORK BEFORE DELIVERY PUMP
082	PIPEWORK BETWEEN DELIVERY PUMP AND FILTER
083	FUEL FILTER
084	FUEL SUPPLY TO INJECTION PUMP
086	LEAK-OFF FUEL LINE
101	TURBOCHARGER (H.P)
111	CHARGE AIR COOLING, IF REQUIRED, LEFT SIDE
113	CHARGE AIR COOLING, RIGHT SIDE
121	AIF FILTER
123	AIR SYSTEM
124	AIR SUPPLY TO CYLINDERS
126	AIR TRANSFER, LEFT - RIGHT SIDE, MAIN PTO END
128	CONTROL BLOCK FOR AIR AND EXHAUST FLOW CONTROL FLAPS
142	CONSTANT-PRESSURE EXHAUST MANIFOLD
143	CARRIER HOUSING
144	EXHAUST FLOW CONTROL FLAP WITH CONTROLS
146	EXHAUST SYSTEM AFTER ENGINE
172	ELECTRIC STARTING
181	LUBE OIL PUMP AND DRIVE
183	OIL FILTRATION/COOLING
184	CRANKCASE-INTERNAL OIL SUPPLY
185	TURBOCHARGER OIL SUPPLY

Overview functional groups

FG FG DESIGNATION

- 188 CONNECTIONS FOR ENGINE OIL SUPPLY
- 202 HIGH-TEMPERATURE CIRCUIT
- 203 LOW-TEMPERATURE CIRCUIT
- 204 HEAT EXCHANGER PLATES
- 206 COOLANT PIPEWORK WITH THERMOSTAT
- 207 RAW WATER PUMPE WITH CONNECTIONS
- 208 COOLANT EXPANSION TANK
- 213 GENERATOR
- 231 ENGINE MOUNTING
- 241 CONNECTION PARTS,ENGINE/GEARBOX,DRIVING END
- 251 STEEL-SPRING COUPLING
- 301 GENERAL ACCESSORIES
- 340 DRIVE, ELECTRIC POWER AND SAFETY SYSTEMS
- 342 DRIVE SYSTEM
- 500 MONITORING AND CONTROL SYSTEM, GENERAL ELECTRICAL EQUIPMENT
- 507 GENERAL WIRING, ENGINE/GEARBOX/PLANT
- 508 ENGINE GOVERNOR (ELECTRONIC) CONTROL SYSTEM
- 511 ALARM SYSTEM
- 517 EMERGENCY INSTRUMENTATION (CONTROL STAND)
- 526 REMOTE CONTROL



00405

INSTALLATION OF LABELS

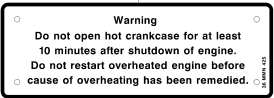
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00410

MISCELLANEOUS MATERIALS (MTN5012)

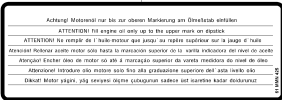
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MISCELLANEOUS MATERIALS (MTN5012)

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MISCELLANEOUS MATERIALS (MTN5012)

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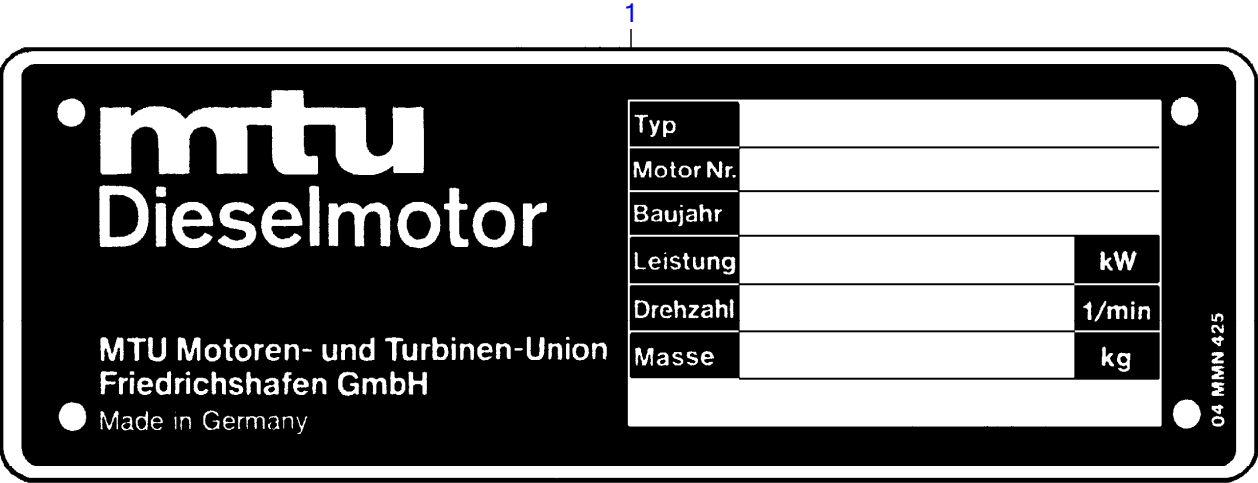
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POS.	PART NO.	DESIGNATION	QTY.	MEASUREMENT
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